Customer Analysis Project Report

Power BI + PySpark ELT Project

1. Project Overview

This project focuses on analyzing customer shopping behavior using Power BI and PySpark-based ELT processes. The goal of the project is to transform raw retail transaction data into meaningful business insights through data engineering, data transformation, and interactive visual analytics.

2. Project Objectives

- Analyze customer purchasing behavior and trends
- Identify top-performing product categories
- Evaluate customer subscription and loyalty behavior
- Study seasonal purchasing patterns
- Measure the impact of discounts and promotions
- Build an interactive Power BI dashboard for business decision-making

3. Dataset Description

Column Name	Description
Customer ID	Unique customer identifier
Age	Customer age
Gender	Gender of customer
Item Purchased	Product purchased
Category	Product category
Purchase Amount (USD)	Transaction amount
Location	Customer location
Season	Season of purchase
Review Rating	Customer review score
Subscription Status	Subscription membership status
Shipping Type	Delivery method
Discount Applied	Discount usage
Promo Code Used	Promo code usage
Payment Method	Payment mode
Frequency of Purchases	Purchase frequency

4. PySpark ELT Process

PySpark was used to perform ELT (Extract, Load, Transform) operations on the raw dataset before visualization in Power BI.

1. Extracted raw CSV dataset
2. Loaded dataset into PySpark DataFrame
3. Performed data cleaning and transformation
4. Handled null values and datatype conversions
5. Standardized categorical values
6. Prepared analytical dataset for reporting
7. Exported transformed data for Power BI visualization

5. Power BI Dashboard Development

The transformed dataset was imported into Power BI for dashboard development. Various interactive visualizations and KPIs were created to analyze customer behavior.

- Sales performance analysis
- Customer demographics analysis
- Seasonal trend analysis
- Subscription behavior analysis
- Discount and promotion analysis
- Shipping and payment insights
- Interactive filters and slicers

6. Key Performance Indicators (KPIs)

KPI	Purpose
Total Revenue	Total sales generated
Average Purchase Amount	Average transaction value
Total Customers	Unique customer count
Average Review Rating	Customer satisfaction analysis
Subscription Conversion Rate	Subscribed customer percentage
Discount Usage Rate	Promotion effectiveness

7. Business Insights

- Certain product categories generated significantly higher revenue.
- Subscription customers showed higher purchase frequency.
- Seasonal trends affected customer buying behavior.
- Discounts and promo codes positively influenced sales.
- Specific payment methods were preferred by customers.

8. Technologies Used

Technology	Purpose
PySpark	ELT and data transformation
Power BI	Dashboard development and visualization

Power Query	Data cleaning
DAX	KPI and measure calculations
CSV Dataset	Source data

9. Conclusion

This project demonstrates the integration of PySpark and Power BI for retail analytics. PySpark was utilized for scalable ELT operations, while Power BI enabled the creation of interactive dashboards and business intelligence reports. The project provides valuable insights into customer purchasing behavior and supports data-driven decision-making.

10. Author

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